

RSPB-Tracker-SMD

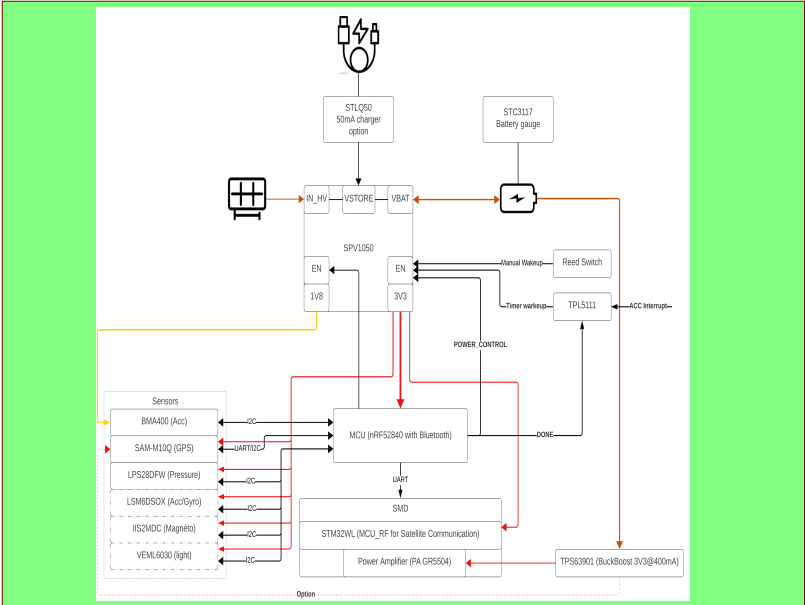
RSPB

The primary objective of the Bird Tracker project is to engineer an low cost, compact, and energy-efficient tracking device based on Argos satellite communication. By adopting the stm32w1 Surface Mount Device (SMD) module, the project endeavors to create a versatile, reusable tracking solution, adaptable for a wide range of species and research applications. The central goal is to deliver a dependable, cost-effective, and highly functional tool that empowers users to precisely track device positions and collect valuable data for diverse research and analytical purposes using state-of-the-art satellite technology. The SMD module could be used alone with doppler localization or combined with a GNSS.

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 - Sheet 15: VEML6030_Light

U_RSPB-Tracker



Legend

General comment such as function title, configuration, ...

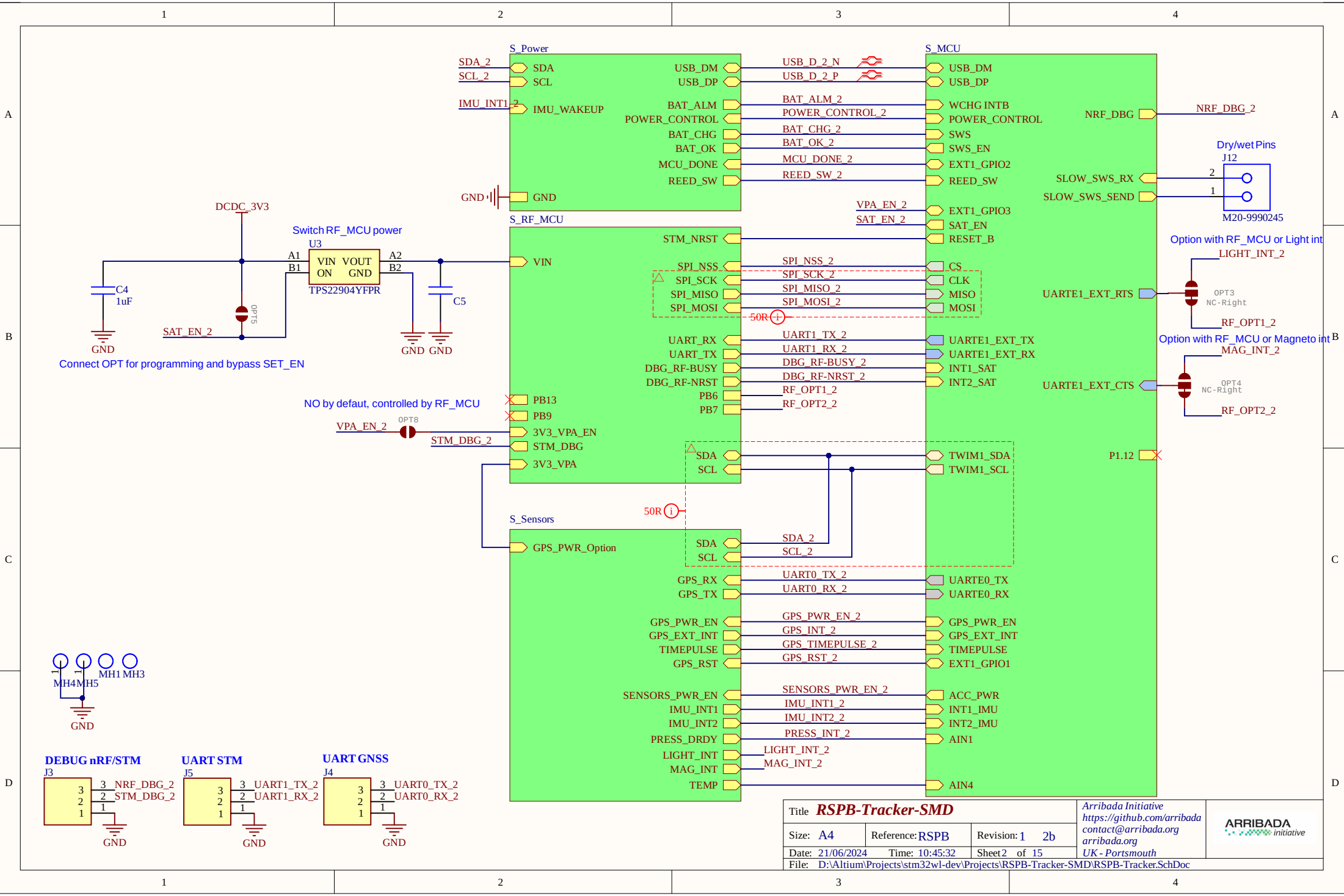
Text to be added to silkscreen.

Warning text.

Notes to generate the board layout.

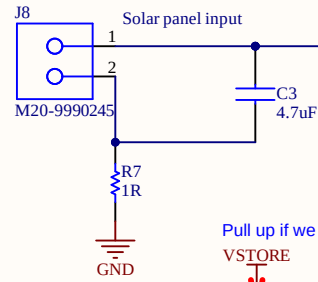
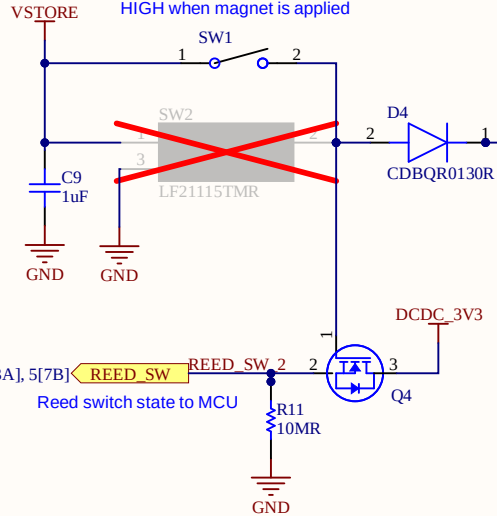
OPEN PLATFORM LICENSE AGREEMENT

Title RSPB-Tracker-SMD			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org UK - Portsmouth		
Size: A4	Reference: RSPB	Revision: 1 2b	Date: 21/06/2024 Time: 10:45:31 Sheet 1 of 15		
File: D:\Altium\Projects\stm32w1-dev\Projects\RSPB-Tracker-SMD\Project_Overview.SchDoc					

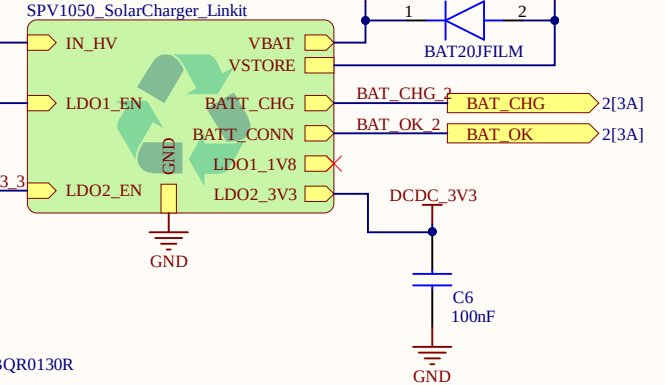


Magnetic switch wakeup

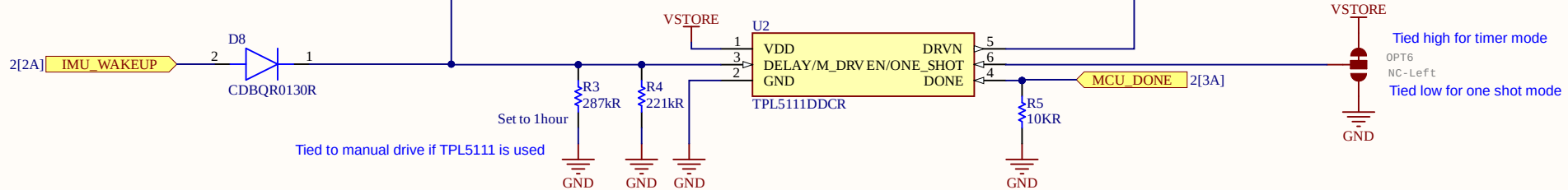
Choice between SW1 or SW2
HIGH when magnet is applied



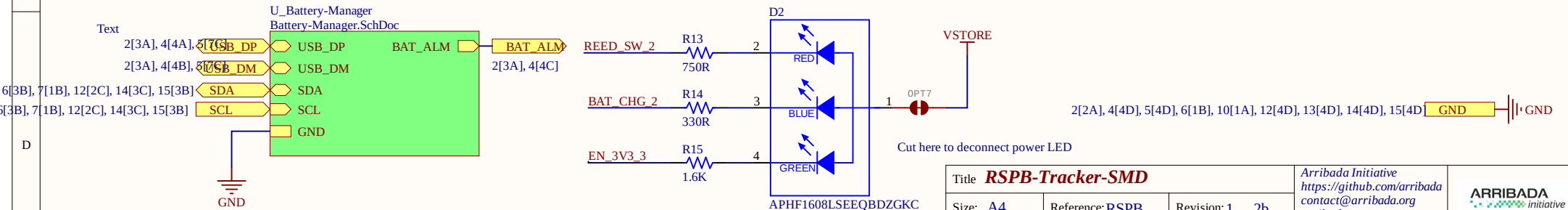
Manage solar panel



Wake up timer low power

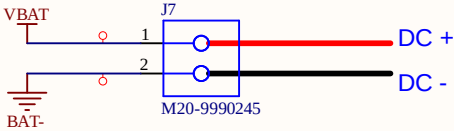


Charge LED (Remove it for operational state)

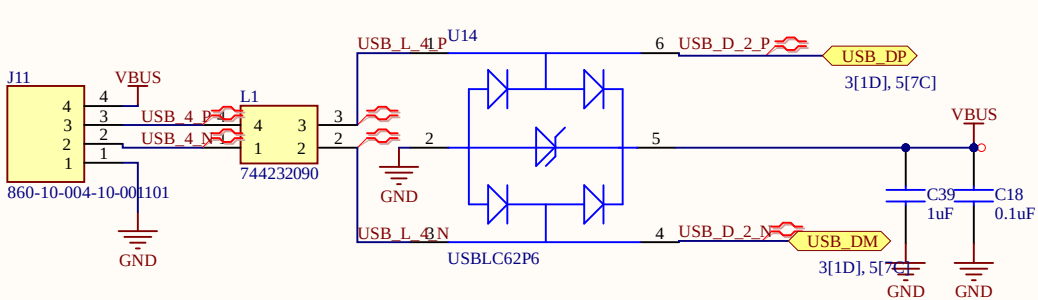


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Size: A4	Reference: RSPB	Revision: 1 2b			
Date: 21/06/2024	Time: 10:45:32	Sheet 3 of 15	UK - Portsmouth		
File: D:\Altium\Projects\stm32wl-dev\Projects\RSPB-Tracker-SMD\Power.SchDoc					

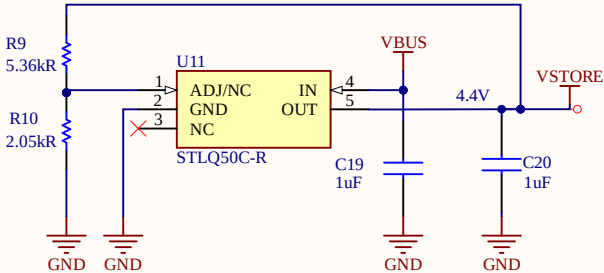
Battery connector



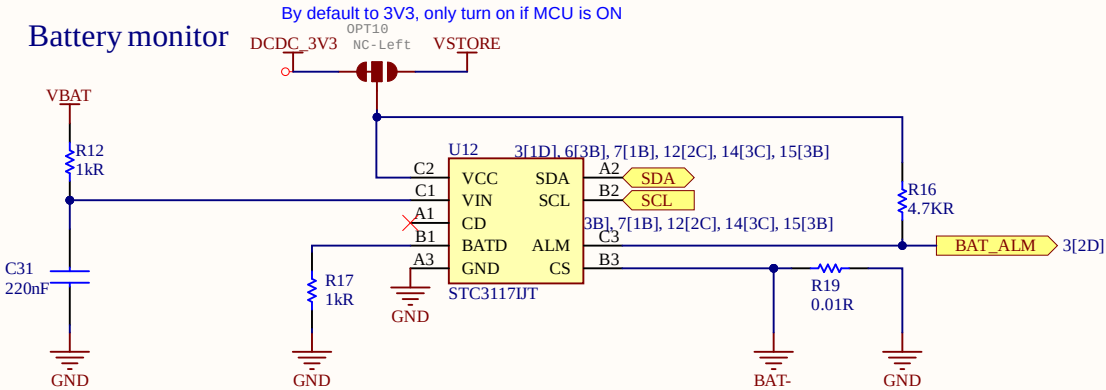
USB header (Optionnal)



USB charger low quiescent current (Optionnal)

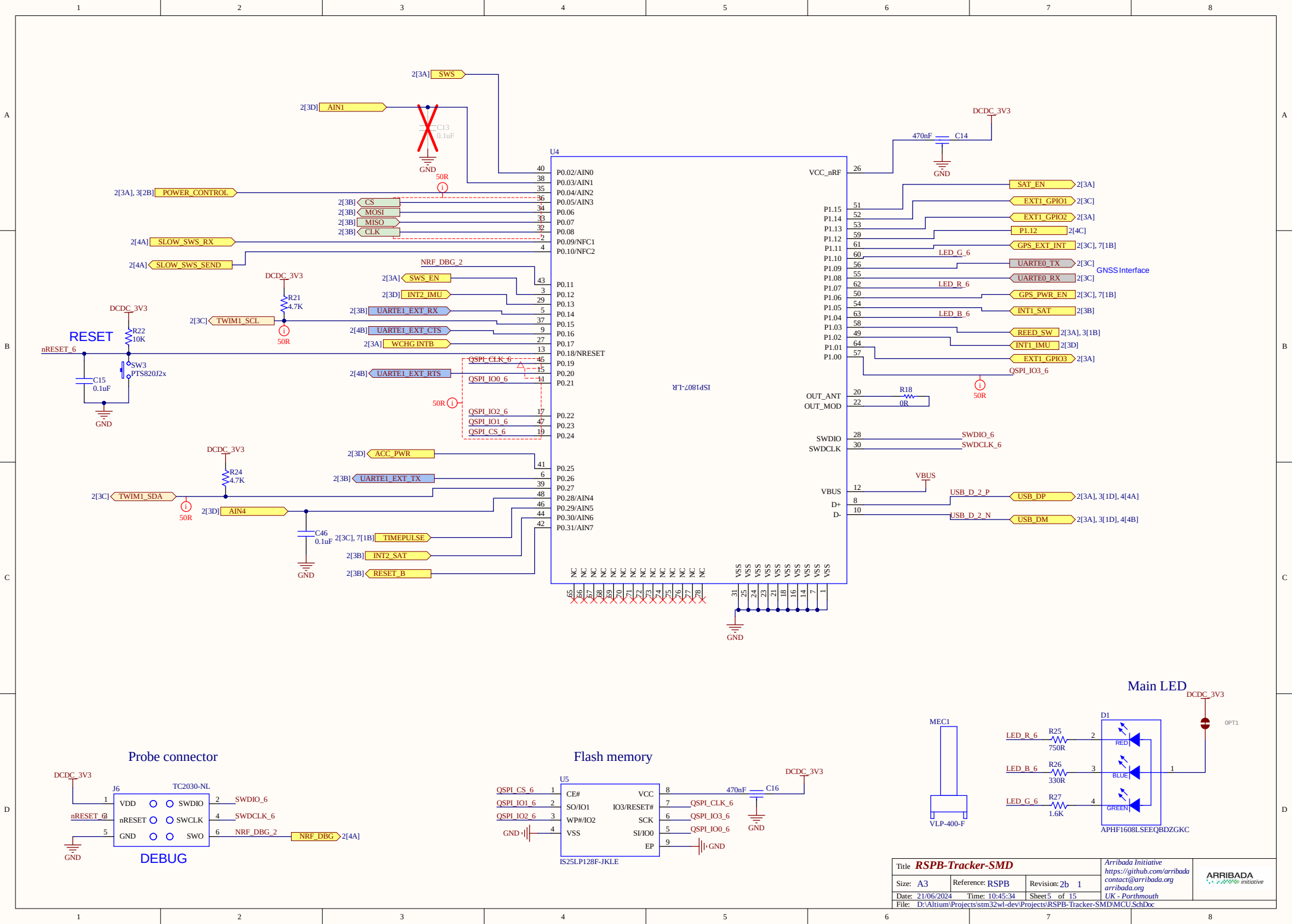


Battery monitor

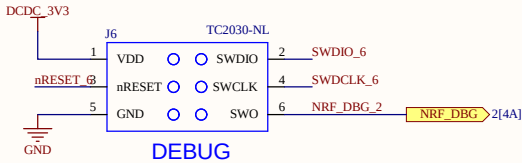


3[1D], 3[4D], 4[4D], 5[4D], 6[1B], 10[1A], 12[4D], 13[4D], 14[4D], 15[4D] GND

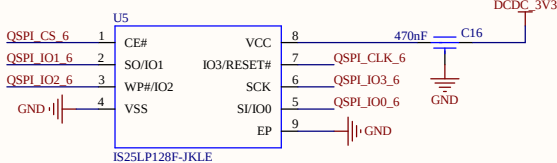
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Size A4	Number	Revision
Date:	6/21/2024	Sheet of
File:	D:\Altium\...\Battery-Manager.SchDoc	Drawn By:



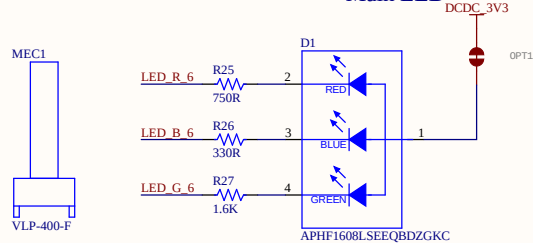
Probe connector

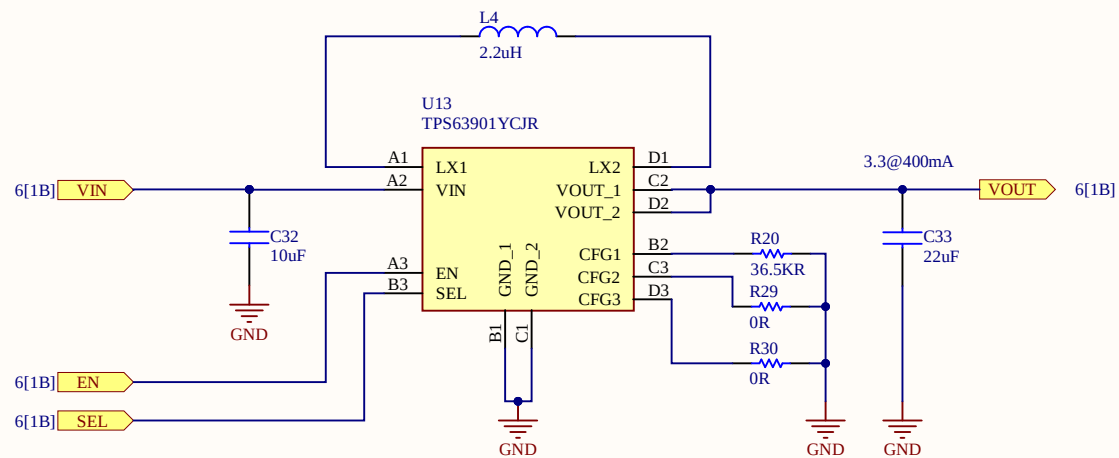


Flash memory



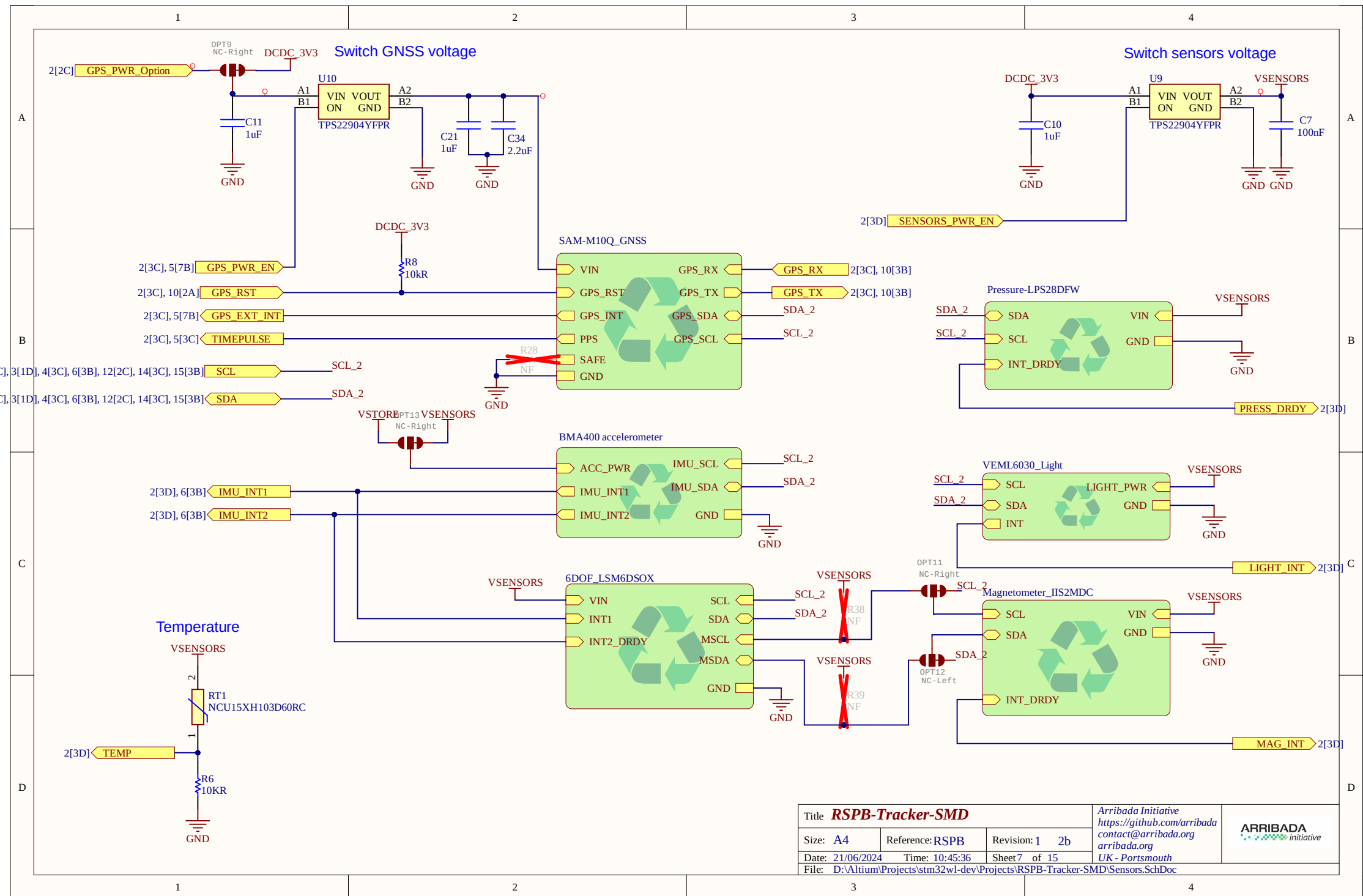
Main LED

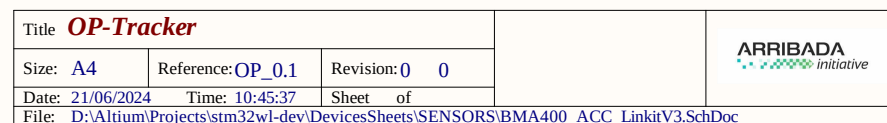




6[1C], 3[4D], 4[4D], 5[4D], 6[1B], 10[1A], 12[4D], 13[4D], 14[4D], 15[4D] GND

Title OP-Tracker			
Size: A4	Reference: OP_0.1	Revision: 0 0	
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File: D:\Altium\Projects\stm32wl-dev\DevicesSheets\Power\TPS63901_BuckBoost_lowIq_3V3_400mA.SchDoc			





1

2

3

4

A

A

B

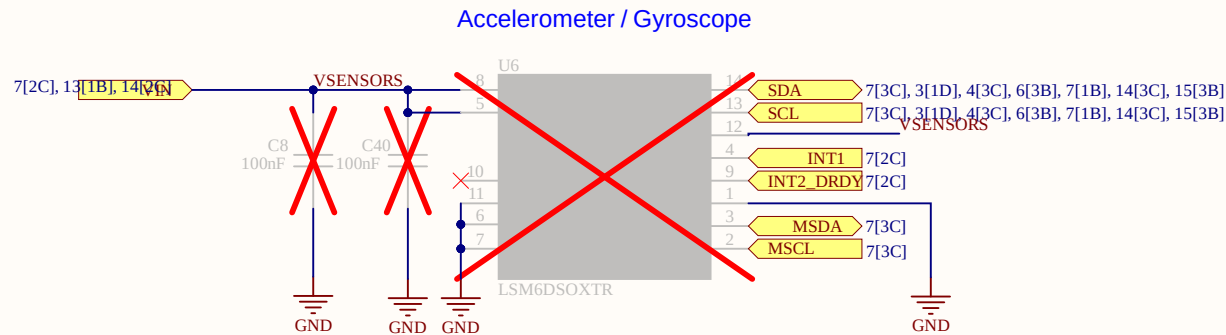
B

C

C

D

D



7[3D], 3[4D], 4[4D], 5[4D], 6[1B], 10[1A], 13[4D], 14[4D], 15[4D] GND

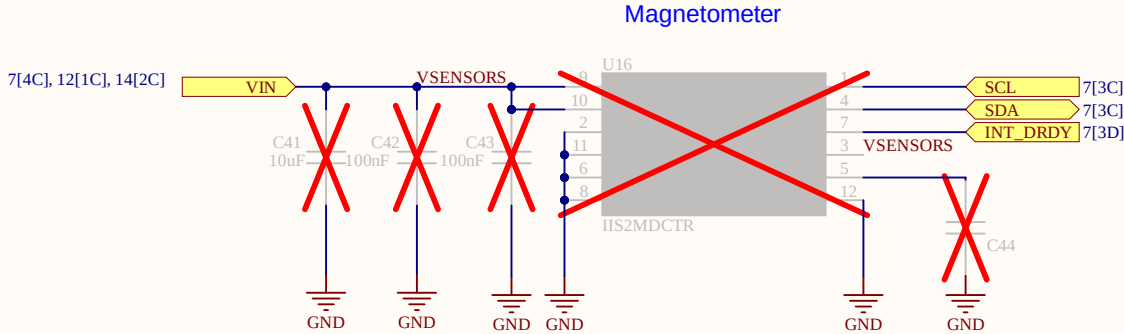
Title RSPB-Tracker-SMD			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org UK - Portsmouth	
Size: A4	Reference: RSPB	Revision: 1 2b		
Date: 21/06/2024	Time: 10:45:37	Sheet 12 of 15		
File: D:\Altium\Projects\stm32wl-dev\DevicesSheets\SENSORS\6DOF_LSM6DSOX.SchDoc				

1

2

3

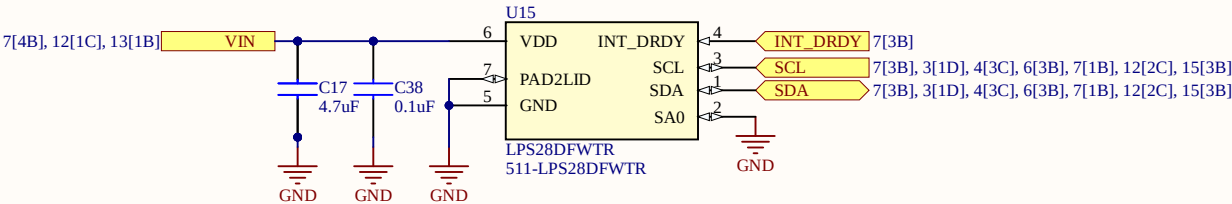
4



7[4C], 3[4D], 4[4D], 5[4D], 6[1B], 10[1A], 12[4D], 14[4D], 15[4D] GND GND

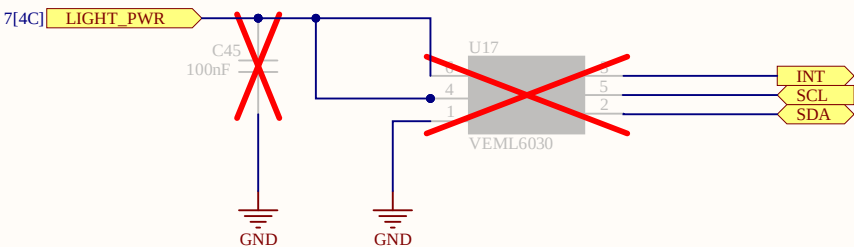
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Size: A4	Reference: RSPB	Revision: 1 2b			
Date: 21/06/2024	Time: 10:45:38	Sheet 13 of 15			
File: D:\Altium\Projects\stm32wl-dev\DevicesSheets\SENSORS\Magnetometer_IIS2MDC.SchDoc					

Pressure Sensor

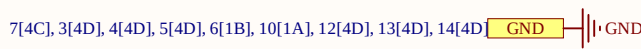


7[4B], 3[4D], 4[4D], 5[4D], 6[1B], 10[1A], 12[4D], 13[4D], 15[4D] GND

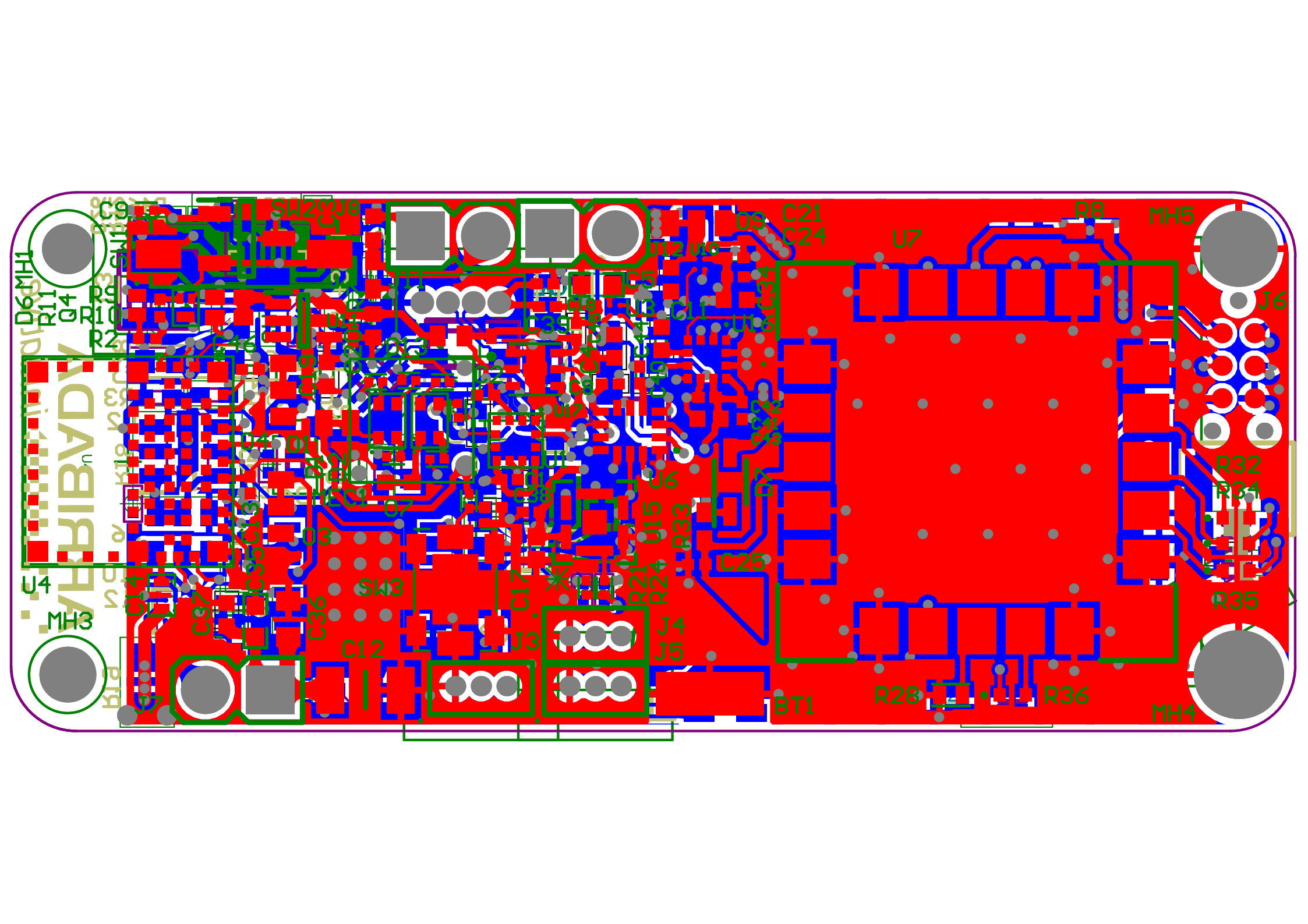
Title RSPB-Tracker-SMD			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org <i>UK - Portsmouth</i>	
Size: A4	Reference: RSPB	Revision: 1 2b		
Date: 21/06/2024	Time: 10:45:38	Sheet 14 of 15		
File: D:\Altium\Projects\stm32wl-dev\DevicesSheets\SENSORS\Pressure-LPS28DFW.SchDoc				



7[3C]
7[3C], 3[1D], 4[3C], 6[3B], 7[1B], 12[2C], 14[3C]
7[3C], 3[1D], 4[3C], 6[3B], 7[1B], 12[2C], 14[3C]



Title RSPB-Tracker-SMD			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org		
Size: A4	Reference: RSPB	Revision: 1	2b		
Date: 21/06/2024	Time: 10:45:39	Sheet 15 of 15			
File: D:\Altium\Projects\stm32wl-dev\DevicesSheets\SENSORS\VEML6030_Light.SchDoc					



Board Stack Report